



CENTRAL EAST EDUCATION DIVISION



2025 MSCE MOCK EXAMINATION BIOLOGY

Wednesday, 26 March

Subject Number: M022/I
Time Allowed: 2 hours
08:00 – 10:00 am

PAPER I (100 marks)

Instructions

1. This paper contains 13 pages. Please Check.
2. Answer all the 13 questions in this paper. The maximum number of marks for each answer is indicated against each question.
3. Write your answers in the spaces provided on the question paper.
4. Write your **name** on top of each page of this paper.
5. In the table provided on this page, tick against the question number you have answered.
6. At the end of the examination, hand in your paper to the invigilators when time is called to stop writing.

Question Number	Tick if answered	Do not write in these columns	
1			
2			
3			
4			
5			
6			
7			
8			
9			
10			
11			
12			
13			

Section A (70 Marks)

Answer **all** the questions in this section in the spaces provided.

1. a. Define reflex arc

(1 mark)

- b. State any **two** examples of reflex actions.

(2 marks)

2. **Figure 1** is a diagram showing a response in potted seedlings exposed to equal amount of light. Use it to answer the questions that follow.

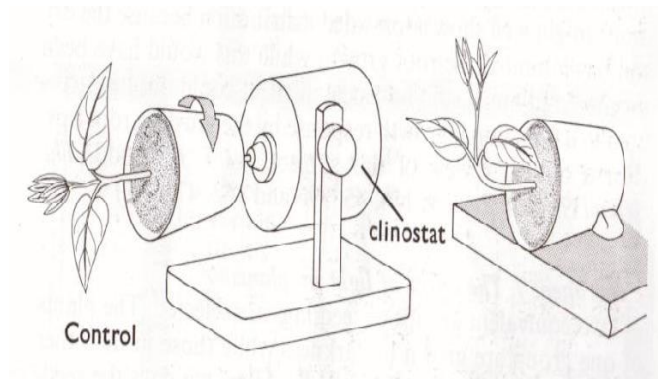


Figure 1

- a (i). Why was the other seedling placed on a clinostat?

(2 marks)

(ii). Explain the results in a seedling not connected to a clinostat.

(3 marks)

b. State the raw material for the dark stage of photosynthesis.

(1 mark)

c. Explain any **one** way in which each of the following leaf structures is adapted for photosynthesis.

(i) Cuticle

(2 marks)

(ii) Lamina

(2 marks)

3. a. State the **two** types of immunity in human body.

(2 marks)

b. Explain how HIV weakens the immune system.

(2 marks)

4. **Figure 2** is a diagram showing blood supply between an embryo and the placenta. Use it to answer the questions that follow.

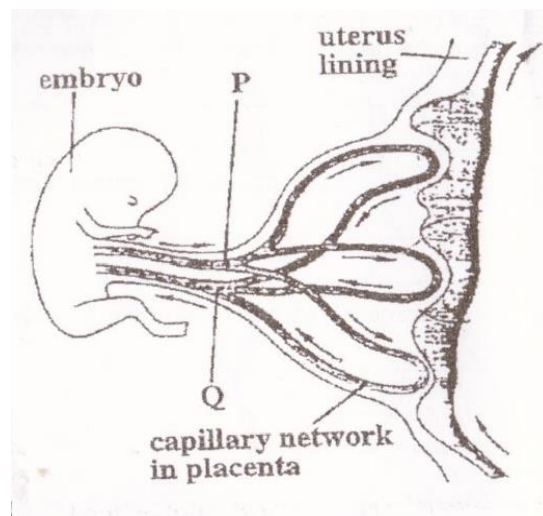


Figure 2

a. (i) Name blood vessels marked **P** and **Q**

P _____ (1 mark)

Q _____ (1 mark)

(ii). State any **two** differences in composition of blood flowing in blood vessels **P** and **Q**

(2 marks)

b. Explain any **two** adaptations of the placenta to its function.

(4 marks)

c. Name **one** organ that starts to function immediately a child is born.

(1 mark)

5. **Figure 3** is a diagram showing cross sections of blood vessels **W** and **Z**.

Use it to answer the questions that follow.

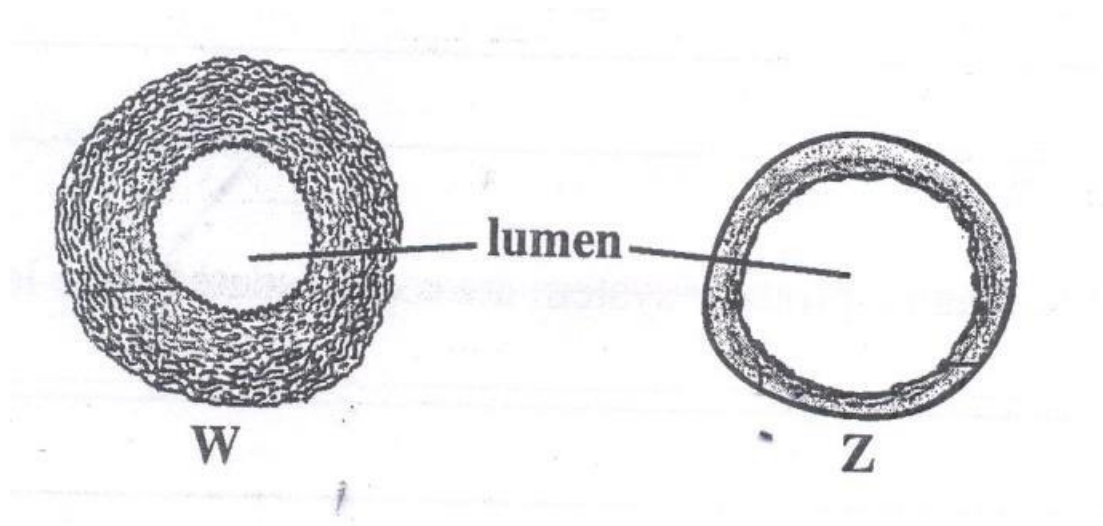


Figure 3

a. State **one** observable difference between the blood vessels **W** and **Z**.

(1 mark)

b. If the two blood vessels were accidentally cut and bleeding occurred, which would be arterial bleeding?

(1 mark)

c. Which of the two blood vessels above contains valves?

(1 mark)

6. a. Define translocation?

(1 mark)

b. Explain any **two** adaptations of the phloem to their functions.

(4 marks)

c. Name the **two** types of vascular bundles.

(2 marks)

d. Explain any **two** ways in which plant hormones are useful in real life.

(4 marks)

7 a. State the importance of the following activities during digestion.

(i) Chewing of food.

(1 mark)

(ii) Protein digestion enzymes are produced in an inactive form.

(1 mark)

b. Mention any **two** nutrients that do not undergo digestion in the bodies of living organisms.

(2 marks)

c. Explain any **two** adaptations of goats to their habitat.

(4 marks)

d. Gregor Mendel crossed pure breeding dwarf garden pea plants with pure breeding tall garden pea plants. The resulting **F₁** generation were all tall.

(i) Using suitable symbols, make a genetic diagram to show the results of mating the **F₁** generation.



(4 marks)

ii. In the above breeding trial involving **F₂** generation, some of the plants are phenotypically tall but heterozygous. How could you distinguish the homozygous tall and those heterozygous?

(2 marks)

8. a. Explain any **two** adaptations of birds for locomotion.

(4 marks)

b. Mention any **two** consequences of mutations.

(2 mark)

9 .a. How is the intercellular fluid formed?

(2 marks)

b. Mention any **two** importance of intercellular fluid in the body.

(2 marks)

10. **Figure 4** is a diagram showing a nerve cell. Use it to answer the questions that Follow.

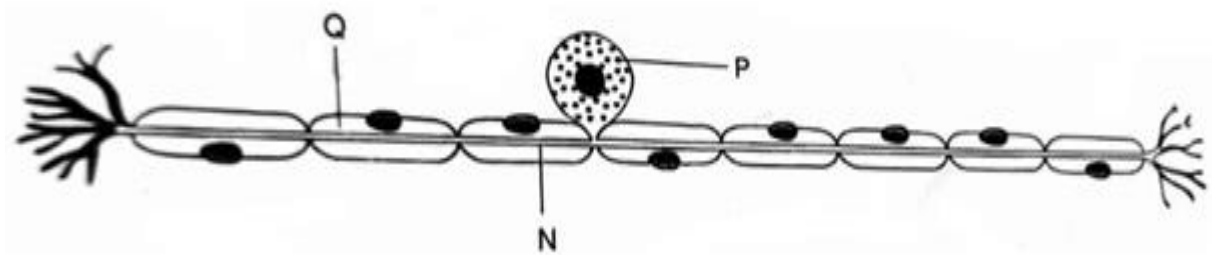


Figure 4

a. (i) Identify the cell

(1 mark)

(ii) Give a reason for your answer in a (i) above.

(1 mark)

b. Name the parts labelled N and Q

N _____

Q _____

(2 marks)

c. State the function of the part labelled Q

(1 mark)

d. Name any neuro transmitter substance found in the synapse.

(1 mark)

Section B (30 marks)

Answer **all** the three questions in this section. Your answers should be in essay form.

11. Explain any **five** factors that could increase the risk of developing cancer in human beings.

[illegible]

(10 marks)

12. Explain any **five** factors to consider before carrying out blood transfusion.

[illegible]

(10 marks)

[illegible]

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